Technical Report

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Introduction

From May 6, 2024 to August 29, 2025, I will have been working at Survalent as a Quality Control intern (QC) for 16 months. The objective of this report is to document the technical skills developed, challenges encountered, and insights gained throughout my internship. It also reflects on how this experience contributed to my academic and professional growth.

Survalent

Survalent is an industry leader in developing software solutions for all types of utilities, including electric, water/wastewater, oil & gas, transit, and renewable energy. With over 60 years in the industry providing its software and services internationally, Survalent has cemented themselves as one of the most trusted Advanced Distribution Management System (ADMS) providers. Survalent's flagship product is the SurvalentONE ADMS that integrates a number of systems and solutions for customers, including a centralized Supervisory Control and Data Acquisition (SCADA) system, Distribution Management System (DMS), Outage Management system (OMS), and many more. Due to how integral the software Survalent provides to utility companies is, attention to detail and ensuring our applications are robust and reliable is imperative when problems may affect millions of people.

Workplace Environment

Survalent has a hybrid work week, where employees are in office Tuesday to Thursday and remote Mondays and Fridays. The software life cycle has a two month release cycle, wherein each team (Dev, QC, e.t.c) will have their own cut off dates per their role in the software life cycle. Each release follows a consistent versioning naming convention where the first number denotes the year, and decimal will separate the release number of the year. For example, QC is currently testing on version 25.4, where ‘25’ indicates this is a 2025 release and ‘.4’ indicates this is the fifth release of this year as indexing starts from 0. This consistent release schedule means that work is fast paced, and collaboration amongst everyone involved is critical in providing a quality product for customers. Many times bugs or issues may be bounced between teams as either the component affected is due to a different team’s changes, or a fix is not as comprehensive and something is still broken. In every case, unilateral communication between every stage and team of the software life cycle is crucial at Survalent.

Survalent, like many other software companies, closely utilizes Git and Team City to manage their versioning. Beyond just the final release versions of each application, there will be multiple builds created for each throughout their development life cycle. These builds will incorporate changes made during that time prior to the final release. Alongside the builds prior to the release version, there is the daily build. Each application has a suite of unit and functional testing that is run through team city, and managed by dedicated developers. Developers will mostly be working with daily build versions, as they are ahead of QC in the software life cycle. This is because the changes they merge into the main branch are what QC has to test.

Internship Role

Quality Control

As a software Quality Control Intern, my role is to identify defects in software products through testing. It involves verifying that the software meets specified requirements and works as intended. QC activities include test planning, test case design, test execution, and bug reporting to ensure the final product is functional and free of critical issues. This means working tightly with product teams to determine specifications of features and confirming what customers will expect from our products, and as well with development teams to ensure those specifications are met and to also test edge cases that the developers may not have had time to test. As well, due to Survalent's strong emphasis on providing quality support, QC works tightly with the support team to both verify specific customer problems, but as well confirm any potential work arounds until a fix can be added to an installation kit provided to the customer. As we work with utility companies in providing critical infrastructure to their management systems that affects millions of people, it is imperative for the software we build to be robust and reliable.

Testing

Testing at Survalent is largely done through black box testing. What this means is that the QC team will test without viewing the source code, or treating the application

like a black box. This means new features and bug fixes are tested by emulating what the end user, usually systems engineers or other related fields, will experience.

Following the two month release schedule, the first month is usually reserved for general Jira tickets assigned to me. As mentioned above, development teams will work ‘ahead’ of QC in the software lifecycle, so that when their changes are merged with the main branch it is ready for QC to test. The second month will be for running regression tests, where larger sweeping tests of applications and features are done.

Jira Tickets

There are two main types of tickets that I test, marked as Development or Bug on Jira. With bugs, what I aim to verify is that the specified issue detailed no longer exists within the application. In the scope of testing for these tickets, black box functional testing is done, where I act similarly to an end user without knowing how the feature was implemented. I test and confirm that the feature is now functioning as expected and the bug no longer exists. Depending on the component fixed, integration testing is also done. For example, if there is a bug with a report generator and another component uses said reports, part of testing the fix would be to see if the second component can accept the reports post-fix.

Development tickets are new features added by developers, and would be tested similarly to bugs. However, instead of confirming that behaviour matches pre-existing documentation and requirements, I discuss either with the Development Analyst (DA) or the Developer assigned to the ticket to confirm the new feature’s requirements. When

the feature is tested, I update the test steps related to that component to now include the newly added feature to ensure comprehensive coverage during regression testing.

Regression Testing

For all the software solutions that Survalent has, each one has a suite of tests that covers a baseline level of functionality. The purpose of running these regression tests is for the sake of sweeping functional and systems testing, confirming that there are no regression issues. When changes are merged into the main branch for a release, there may be components that were working fine prior to a new feature being added, but breaks unexpectedly after merging. These are called regression issues that the manual regression test executions are meant to find.

These tests are not static either, and are maintained and updated by everyone on the QC team to ensure that they accurately and comprehensively cover key functionalities. As mentioned above, each Dev ticket tested will warrant an update to the test suite related to that component, as a new feature would need new test steps. Beyond just new features, general upkeep of test steps is important as well to ensure clarity for new testers. Due to how varied the applications are at Survalent, QC members will regularly ramp up tests for components they are not usually working with to ensure that everyone will at least understand something about the other applications.

Projects

Over my 16 months working at Survalent, I have had the opportunity to work on many projects and software. This ranges from working closely with development teams, to test automation and to the general day to day testing that I do.

SCADA

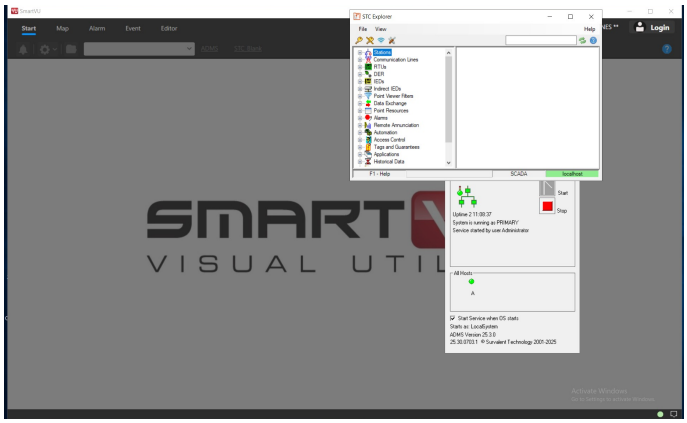


Fig 1. Screenshot of Scada Tools. Taken by author

The general regression testing and bugs that I was assigned were mostly with the SCADA core features, including STC Explorer, SmartVu and ADMS Server applications. ADMS Server acts as an independent server for all the SCADA applications to communicate with. It can be configured with up to four redundant servers to ensure up time even if one were to fail. STC Explorer is the main interface for systems engineers to maintain their system. This provides tools such as user rights, station management, command sequencing, e.t.c. SmartVu is the graphical side of things, where a map can be overlaid with the utility company's coverage, visually representing any changes in the grid. Alarms will activate based on outages, and depending on the customer's organization, it could allow controllers to remotely turn off specific switches or even dispatch technicians to go in person.

Bug and Dev ticket testing will vary quite wide even within the individual applications. Tests fixes for a ui element, to a specific set of actions causing the application to crash. Regression testing varies just as much. One of the common test suites are base line tests, that sweep over certain core features of larger applications. This is to ensure that everything is at least functional. I've also tested licensing, specific utilities such as UtlTools, point viewers, SmartVu's control paneling and line editors, e.t.c.

GIS Wizard

GIS Wizard is an application developed by Survalent to integrate already existing map data from control rooms with Survalent's own ADMS systems. Other features include data validation, importing map data, and a number of other critical spatial data controls. As one of the two QC members dedicated to this development team, we test and validate any changes and bug fixes the GIS Wizard team makes. Recently as well, they are migrating from a desktop windowed application to a web application. This comes with new test suites that have to be written to accommodate these changes. As well, being a new project it is relatively unstable and features are constantly being added and worked on. This means this team's work has become very fast paced with tickets being created, merged, and brought to testing in a short turnaround time.

Automation

The majority of testing done is manual, where QC members will proceed through test steps and verify functionality at every step. Certain tests suites are candidates for

automation wherein the tests are repetitive in nature, or potentially could be done via an automation tool. Certain tests were automated already, but do not provide full coverage. For some time I was managing a number of these automation suites and updating them to match current needs. The suites I managed were built with Ranorex's remote spy agent dedicated to desktop windowed applications. However due to the time expenditure in comparison to overall coverage of the automated tests, a different approach to automation was taken.

Technical Knowledge

Throughout my time at Survalent, I developed a lot of my skills. Notably, I learned a lot about project design and planning. This is because as QC, a lot of my day to day tasks involve confirming the quality of an application's end design and its efficacy. This allows me to really key in on the importance of proper design. Being able to sit in with development teams, being involved in discussions on how new features are developed, and working tightly with teams in all stages of the software life cycle has provided me a unique opportunity to learn how software features are built from the initial planning stage, all the way to testing and shipping. This is especially true when I was tasked with writing test steps for new features and even new applications entirely. Balancing coverage of testing while also considering time investment of manual testing in a tight two month release cycle was very critical. I learned a lot in terms of project planning and considering things from a customer's perspective and what would be important for them.

As well, due to the nature of black box testing, I learned a lot in terms of ADMS and SCADA systems, especially in the context of how Utility Companies work with them. Working at Survalent allowed me to learn more of an industry that I otherwise knew nothing about.

Other technical aspects I learned while working at Survalent include, networking principles, web application testing, Ranorex automated test development, GIS Systems and how large scale git version control is handled across multiple applications. Some other skills that I further developed were SQL querying, Bash scripting, and functional testing.

Reflection

Expectations vs. Reality

Before working at Survalent, I did not have a complete understanding of how intensive software testing could be. In my past internships and with my own projects, there were no dedicated test teams. Functional and unit testing was done on a much more surface level before software was shipped. Working at Survalent has really helped me develop a new found appreciation into software testing.

Future Goals

Prior to working at Survalent, my career goals were to work in software development. Because of how broad software can be, it felt almost like choice paralysis. I could go into full stack development, working with back to back solutions, pure front end or web development and focus on a design skill set, and a dozen other types of

software development. However, Testing and Quality Control in general was a field that I did not have any experience in working with at all. After working at Survalent, I found that testing is a lot more interesting than I initially thought. Granted, some test suites can be very repetitive and menial, but finding bugs and working closely with product teams and developers is a very rewarding experience. Planning out and writing test steps was also surprisingly interesting, as it was about striking a balance between test coverage and the time investment. For example, not every pixel of an application needs to be tested, but instead sweeping validation that all the ui elements are showing up at all is a much more efficient means of testing. As well having to think beyond just the usual inputs for an application or feature, considering corner cases and integration issues for testing was very interesting.

Working in quality control requires an unique perspective between the technical side of development and the higher level overview that customers would see. Although I am not sure if I would like to still pursue pure QC as a career choice, I am considering testing a lot more seriously after my time at Survalent. After some research, many software companies have ‘Software Engineers in Test’, where their main responsibilities include writing test cases for applications, either through automation or unit testing. This is more of a blend between software development and testing that I did not know existed prior to working in QC. I still am not set on a career path, but this internship has opened my eyes to an entirely new aspect of the software development lifecycle.

Relevant School Work

The project development courses, 2212 and 3307 were very helpful in priming me for the software life cycle and collaborative work. This is especially true with Survalent's fast paced two month release cycle, where everything I learned in those classes were put into practice.

School

Working for 16 months has really given me a different perspective on developing a work life balance. Having that block of 9-5 dedicated to just work really puts into perspective how much time I have in a day, and how to be more efficient in my time management. Balancing work, gym, social events and my own side projects was really challenging at first, but I learned a lot of techniques to do so. I am worried I won't be used to school life again after being away for so long, but I believe the time management skills I developed over this internship will help me greatly in the coming year.